

THE HERITAGE SCHOOL
FIRST TERM EXAMINATION, 2026-27
MATHEMATICS · CLASS 12 · PRACTICE PAPER 3

Maximum Marks: 80

Time Allotted: Three Hours

Instructions to Candidates

- You are allowed an additional 15 minutes for only reading the paper.
- You must NOT start writing during reading time.
- The Question Paper consists of **twenty questions** and **four sections**: Sections A, B, C and D. **All questions are compulsory.**
- **Section A** consists of very short answer questions of **1 mark** each.
- **Section B** consists of short answer questions of **2 marks** each.
- **Section C** consists of moderately long answer questions of **3 marks** each.
- **Section D** consists of long answer questions of **5 marks** each.
- Internal choices have been provided in **three questions** in each of Sections B, C and D.
- While attempting Multiple Choice Questions in Section A, you are required to write only ONE option as the answer.
- The intended marks for questions or parts of questions are given in brackets [].
- All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
- Mathematical tables and graph papers are provided.

Syllabus covered: Relations and Functions; Inverse Trigonometric Functions; Matrices; Determinants; Continuity and Differentiability; Applications of Derivatives; Integrals; Application of Integrals; Linear Programming.

Not examined in this paper (held back to the Pre-Board): Differential Equations, Vectors, Three-dimensional Geometry, Probability.

SECTION A - 20 MARKS

Question 1 [20 × 1]

In sub-parts (i) to (xx), choose the correct option or answer the question as instructed.

- (i) The number of equivalence relations on the set $\{1, 2, 3\}$ that contain the pair $(1, 2)$ is:
- (a) 2
(b) 1
(c) 3
(d) 5
- (ii) The range of the function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x^2 + 1$ is:
- (a) $[1, \infty)$
(b) $\mathbb{R}$
(c) $(1, \infty)$
(d) $(0, \infty)$
- (iii) Find the value of $\cos^{-1}(\cos 7\pi/6)$.
- (iv) If $\sin^{-1}x = \pi/5$, write the value of $\cos^{-1}x$.
- (v) If $A = [a_{ij}]$ is a symmetric matrix, state the relation between a_{12} and a_{21} .
- (vi) Find the number of all possible 2×2 matrices whose entries are 0 or 1.
- (vii) If A is a 2×2 matrix with $|A| = 5$, find $|A \cdot \text{adj } A|$.
- (viii) Write the minor of the element 4 in the determinant

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

- (ix) At which point is $f(x) = |x - 3|$ not differentiable?
- (x) Evaluate $d/dx(\sin^{-1}x + \cos^{-1}x)$.

(xi) If $y = x^5$, find $(d^2y)/(dx^2)$.

(xii) If $x = at$ and $y = a/t$, find dy/dx .

(xiii) Write the equation of the tangent to the curve $y = x^2$ at the point $(1, 1)$.

(xiv) The function $f(x) = \cos x$ is strictly decreasing on:

- (a) $(0, \pi)$
- (b) $(-\pi, 0)$
- (c) $(0, \pi/2)$ only
- (d) no interval

(xv) Evaluate $\int \tan x \, dx$.

(xvi) Evaluate $\int_0^1 x^2 \, dx$.

(xvii) Evaluate $\int_{-2}^2 x^5 \, dx$.

(xviii) Find the area bounded by $y = 4x$, the x -axis and the lines $x = 0$, $x = 2$.

(xix) The corner points of a feasible region are $(0,0)$, $(3,0)$, $(2,2)$ and $(0,3)$. Find the maximum value of $Z = x + y$.

(xx) **Assertion (A):** $\int_{-a}^a f(x) \, dx = 0$ whenever f is an odd function.

Reason (R): a function f is odd if $f(-x) = -f(x)$ for every x in the domain.

- (a) Both A and R are true and R is the correct explanation of A
- (b) Both A and R are true but R is not the correct explanation of A
- (c) A is true but R is false
- (d) A is false but R is true

SECTION B - 14 MARKS

Question 2 [2]

If $f(x) = 2x + 3$ and $g(x) = x^2$, find $(fog)(2)$ and $(gof)(2)$.

Question 3 [2]

Find the value of $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$.

OR

Simplify $\cos^{-1} \sqrt{(1+x)/2}$ for $0 < x < 1$.

Question 4 [2]

Find x and y if

$$2 \begin{bmatrix} 1 & 3 \\ 0 & x \end{bmatrix} + \begin{bmatrix} y & 0 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 5 & 6 \\ 1 & 8 \end{bmatrix}$$

Question 5 [2]

Using determinants, find the area of the triangle with vertices $(-2, -3)$, $(3, 2)$ and $(-1, -8)$.

OR

Find dy/dx if $x^{2/3} + y^{2/3} = a^{2/3}$.

Question 6 [2]

Differentiate $\log(\sin(x^2))$ with respect to x .

Question 7 [2]

Evaluate $\int (\sin^{-1} x)/(\sqrt{1-x^2}) \, dx$.

Question 8 [2]

Find the local maximum value of $f(x) = x^3 - 3x$.

OR

Find the slope of the normal to the curve $x = a \cos^3 \theta$, $y = a \sin^3 \theta$ at $\theta = \pi/4$.

SECTION C - 21 MARKS

Question 9 [3]

Show that the relation R on $\mathbb{Z}$ defined by $R = \{(a, b) : 2 \text{ divides } (a - b)\}$ is an equivalence relation, and write its equivalence classes.

Question 10 [3]

Using properties of determinants, prove that

$$\begin{vmatrix} a+b+2c & a & b \\ c & b+c+2a & b \\ c & a & c+a+2b \end{vmatrix} = 2(a+b+c)^3$$

OR

Find a and b so that

$$\begin{vmatrix} 5 & x \leq 2 \\ ax+b & 2 < x < 10 \\ 21 & x \geq 10 \end{vmatrix}$$

is continuous everywhere.

Question 11 [3]

Find the equations of the tangent and the normal to the curve $x^{2/3} + y^{2/3} = 2$ at the point $(1, 1)$.

OR

Differentiate $\sin^{-1}(2x\sqrt{1-x^2})$ with respect to x , for $|x| < 1/\sqrt{2}$.

Question 12 [3]

A ladder 5 m long leans against a wall. Its foot is pulled away from the wall at 2 m/s. How fast is the top sliding down the wall when the foot is 3 m from the wall?

OR

Find the intervals in which $f(x) = x^3 - 4x^2 + 4x + 15$ is strictly increasing.

Question 13 [3]

Evaluate $\int (x+2)/(\sqrt{x^2+2x+3}) dx$.

Question 14 [3]

Evaluate $\int_0^{\pi/4} \tan^2 x dx$.

Question 15 [3]

Solve graphically: maximise $Z = 4x + y$ subject to $x + y \leq 50$, $3x + y \leq 90$, $x \geq 0$, $y \geq 0$.

SECTION D - 25 MARKS

Question 16 [5]

Show that $f : [-1, 1] \rightarrow \mathbb{R}$, $f(x) = x/(x+2)$ is one-one, and find the inverse of f regarded as a map onto its range.

Question 17 [5]

Solve the following system of equations by the matrix method:

$$2x + 3y + 3z = 5, \quad x - 2y + z = -4, \quad 3x - y - 2z = 3$$

Question 18 [5]

A tank with a rectangular base and rectangular sides, open at the top, is to be constructed so that its depth is 2 m and its volume is 8 m^3 . If building the base costs ₹70 per m^2 and the sides ₹45 per m^2 , find the least cost of construction.

OR

Show that the right circular cone of least curved surface area and given volume has an altitude equal to $\sqrt{2}$ times the radius of the base.

Question 19 [5]

Evaluate $\int \sqrt{3 - 2x - x^2} \, dx$.

OR

Evaluate $\int (e^x(1 + x))/(\cos^2(x e^x)) \, dx$.

Question 20 [5]

Find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 16$.

OR

Find the area of the region enclosed between the curves $y = x^2$ and $y = 2 - x^2$.

ANSWER KEY & MARKING NOTES

Practice Paper 3. **Do not read this section before attempting the paper.** Answers were checked against a computer algebra system; the notes give the decisive step rather than a full solution.

Section A

Q	Answer
1 (i)	(a) 2. Such a relation must contain (2,1) as well; the two possibilities are $\{(1,1),(2,2),(3,3),(1,2),(2,1)\}$ and the universal relation on the set.
1 (ii)	(a) $[1, \infty)$. $x^2 \geq 0$ with equality at $x = 0$, so $f(x) \geq 1$ and the value 1 is attained.
1 (iii)	$5\pi/6$. $7\pi/6 \notin [0, \pi]$, but $\cos 7\pi/6 = \cos(2\pi - 7\pi/6) = \cos 5\pi/6$.
1 (iv)	$3\pi/10$. Since $\sin^{-1}x + \cos^{-1}x = \pi/2$, $\cos^{-1}x = \pi/2 - \pi/5$.
1 (v)	$a_{12} = a_{21}$. Symmetry means $A' = A$, i.e. $a_{ij} = a_{ji}$ for all i, j .
1 (vi)	16. Each of the four entries is chosen in 2 ways, so $2^4 = 16$.
1 (vii)	25. $A \cdot \text{adj } A = A I_2 = 5I_2$, whose determinant is $5^2 = 25$.
1 (viii)	1. Deleting the row and column through 4 leaves the single entry 1.
1 (ix)	$x = 3$. The graph has a corner there; the left and right derivatives are -1 and 1 .
1 (x)	0. The sum is the constant $\pi/2$ for every x in $[-1, 1]$.
1 (xi)	$20x^3$.
1 (xii)	$-1/t^2$. $dy/dx = (-a/t^2)/a = -1/t^2$.
1 (xiii)	$y = 2x - 1$. The slope is $dy/dx = 2$ at $x = 1$.
1 (xiv)	(a) $(0, \pi)$. $f'(x) = -\sin x < 0$ throughout $(0, \pi)$.
1 (xv)	$\log \sec x + C$, equivalently $-\log \cos x + C$.
1 (xvi)	$1/3$. $[x^3/3]_0^1 = 1/3$.
1 (xvii)	0. x^5 is odd and the interval is symmetric about the origin.
1 (xviii)	8 square units. $\int_0^2 4x \, dx = [2x^2]_0^2 = 8$.
1 (xix)	4, at $(2, 2)$. The values are 0, 3, 4 and 3.
1 (xx)	(a). Splitting the integral at 0 and substituting $x \rightarrow -x$ in the left half makes the two halves cancel, and it is exactly the oddness in R that produces the cancellation.

Section B

Q	Answer
2	$(f \circ g)(2) = f(g(2)) = f(4) = \mathbf{11}$; $(g \circ f)(2) = g(f(2)) = g(7) = \mathbf{49}$. Note $f \circ g \neq g \circ f$.
3	Combine the last two: $2 \cdot 3 = 6 > 1$, so $\tan^{-1} 2 + \tan^{-1} 3 = \pi + \tan^{-1}((2+3)/(1-6)) = \pi + \tan^{-1}(-1) = \pi - \pi/4 = 3\pi/4$. Adding $\tan^{-1} 1 = \pi/4$ gives $\mathbf{\pi}$.
3 (OR)	Put $x = \cos 2\theta$, so $\theta = (1/2)\cos^{-1}x$. Then $\sqrt{((1 + \cos 2\theta)/2)} = \sqrt{(\cos^2 \theta)} = \cos \theta$, and the expression is $\cos^{-1}(\cos \theta) = \theta = \mathbf{(1/2)\cos^{-1}x}$.
4	The left side is $[[2 + y, 6], [1, 2x + 2]]$. Comparing entries, $2 + y = 5$ and $2x + 2 = 8$, so $\mathbf{x = 3}$ and $\mathbf{y = 3}$.
5	Area = $(1/2) \det[[-2, -3, 1], [3, 2, 1], [-1, -8, 1]] = (1/2) -30 = \mathbf{15 \text{ square units}}$.
5 (OR)	Differentiating, $(2/3)x^{-1/3} + (2/3)y^{-1/3} \cdot dy/dx = 0$, so $\mathbf{dy/dx = -(y/x)^{1/3}}$.
6	By the chain rule, $(2x \cos(x^2))/(\sin(x^2)) = \mathbf{2x \cot(x^2)}$.
7	Put $t = \sin^{-1}x$, so $dt = dx/\sqrt{1-x^2}$ and the integral is $\int t \, dt = \mathbf{((\sin^{-1}x)^2)/2 + C}$.
8	$f'(x) = 3x^2 - 3 = 0$ gives $x = \pm 1$. $f''(x) = 6x$, which is negative at $x = -1$, so there is a local maximum there. Local maximum value $f(-1) = 2$.

Q	Answer
8 (OR)	$dy/dx = (3a \sin^2 \theta \cos \theta)/(-3a \cos^2 \theta \sin \theta) = -\tan \theta$, which is -1 at $\theta = \pi/4$. The normal has slope 1 .

Section C

Q	Answer
9	Reflexive: $2 \mid 0$. Symmetric: if $a - b = 2k$ then $b - a = -2k$. Transitive: $a - b = 2k$ and $b - c = 2m$ give $a - c = 2(k + m)$. Two integers are related exactly when they have the same parity. Classes: the even integers and the odd integers.
10	Apply $C_1 \rightarrow C_1 + C_2 + C_3$: every entry of the first column becomes $2(a + b + c)$. Take that factor out, then use $R_2 \rightarrow R_2 - R_1$ and $R_3 \rightarrow R_3 - R_1$ to reach a triangular form, producing $(a + b + c)^2$. Verified symbolically.
10 (OR)	$2a + b = 5$ and $10a + b = 21$, so $8a = 16$ and $a = 2, b = 1$.
11	$dy/dx = -(y/x)^{1/3} = -1$ at $(1, 1)$. Tangent: $x + y = 2$. Normal: $y = x$.
11 (OR)	Put $x = \sin \theta$; then $2x\sqrt{1 - x^2} = \sin 2\theta$, and for $ x < 1/\sqrt{2}$ the expression equals $2 \sin^{-1} x$. Hence the derivative is $2/(\sqrt{1 - x^2})$.
12	$x^2 + y^2 = 25$ gives $x(dx/dt) + y(dy/dt) = 0$. At $x = 3, y = 4$, so $3(2) + 4(dy/dt) = 0$ and $dy/dt = -3/2$: the top slides down at $3/2$ m/s.
12 (OR)	$f(x) = 3x^2 - 8x + 4 = (3x - 2)(x - 2)$. Strictly increasing on $(-\infty, 2/3)$ and $(2, \infty)$.
13	Write $x + 2 = (1/2)(2x + 2) + 1$ and $x^2 + 2x + 3 = (x + 1)^2 + 2$. The two parts give $\sqrt{x^2 + 2x + 3} + \log (x + 1) + \sqrt{x^2 + 2x + 3} + C$.
14	$\tan^2 x = \sec^2 x - 1$, so the integral is $[\tan x - x]_0^{\pi/4} = \mathbf{1 - \pi/4}$ (≈ 0.2146).
15	Corner points $(0,0), (30,0), (20,30)$ and $(0,50)$ give $Z = 0, 120, 110$ and 50 . Maximum $Z = 120$ at $(30, 0)$.

Section D

Q	Answer
16	One-one: $x_1/(x_1 + 2) = x_2/(x_2 + 2)$ cross-multiplies to $2x_1 = 2x_2$, so $x_1 = x_2$. Solving $y = x/(x + 2)$ gives $x(1 - y) = 2y$, i.e. $x = 2y/(1 - y)$. $f^{-1}(y) = 2y/(1 - y)$.
17	$A = \begin{bmatrix} 2 & 3 & 3 \\ 1 & -2 & 1 \\ 3 & -1 & -2 \end{bmatrix}$ with $ A = 40 \neq 0$, so $X = A^{-1}B$ gives $x = 1, y = 2, z = -1$. Check: $2 + 6 - 3 = 5 \checkmark, 1 - 4 - 1 = -4 \checkmark, 3 - 2 + 2 = 3 \checkmark$.
18	Depth 2 and volume 8 force a base area of 4, so if the base is $x \times y$ then $xy = 4$. Cost $C = 70(4) + 45 \cdot 2 \cdot 2(x + y) = 280 + 180(x + 4/x)$. $dC/dx = 180(1 - 4/x^2) = 0$ gives $x = 2$, hence $y = 2$, and $(d^2C)/(dx^2) > 0$ there. Least cost = ₹1000.
18 (OR)	With $V = (1/3)\pi r^2 h$ fixed, the curved surface is $S = \pi r\sqrt{r^2 + h^2}$. Minimising S^2 subject to the volume constraint gives $h^2 = 2r^2$, i.e. $h = \sqrt{2} r$.
19	Complete the square: $3 - 2x - x^2 = 4 - (x + 1)^2$. Using $\int \sqrt{a^2 - t^2} dt = (t/2)\sqrt{a^2 - t^2} + (a^2/2) \sin^{-1}(t/a)$ with $t = x + 1, a = 2$, the answer is $((x + 1)/2)\sqrt{3 - 2x - x^2} + 2 \sin^{-1}((x + 1)/2) + C$.
19 (OR)	Put $t = x e^x$, so $dt = e^x(1 + x) dx$ and the integral becomes $\int \sec^2 t dt = \mathbf{\tan(x e^x) + C}$.
20	The line meets the circle where $2x^2 = 16$, i.e. $x = 2\sqrt{2}$. Area = $\int_0^{2\sqrt{2}} x dx + \int_{2\sqrt{2}}^4 \sqrt{16 - x^2} dx = 4 + (2\pi - 4) = \mathbf{2\pi \text{ square units}}$ (≈ 6.28), which is one eighth of the disc.
20 (OR)	$x^2 = 2 - x^2$ gives $x = \pm 1$, and $2 - x^2$ is the upper curve between them. Area = $\int_{-1}^1 (2 - x^2) dx = \mathbf{8/3 \text{ square units}}$.